

Pre-Requisites

To effectively participate in this session, learners are required to:

- Awareness of what NLP means in simple terms (machines working with human language).
- Basic familiarity with business problems that involve text (customer feedback, support chats, compliance docs).
- Awareness of the NLP pipeline:
data → preprocessing → feature extraction → modeling → evaluation → business impact

Session Summary

This 2-hour live session focuses on exploring **two real-world NLP industry case studies** using traditional approaches. Learners will see how different organizations translated business challenges into NLP tasks, applied the classical NLP pipeline (data preprocessing, feature extraction, and machine learning models), and derived actionable insights. The instructor will guide learners through both case studies step by step, highlighting problem statements, solution approaches, results, and business impact.

Session-related slides and reference materials for the instructor can be found in the [this](#) folder.

Note: This is a case study-driven session. Learners are encouraged to actively participate in Q&A segments and a guided discussion where they share domain-relevant examples and reflect on how similar NLP workflows can be applied in their own contexts.

Learning Objectives/Outcomes

By the end of this session, learners will be able to:

- Describe the step-by-step flow of a traditional NLP pipeline in industry settings
- Explain how real-world business problems can be reframed as NLP tasks
- Analyze two case studies to understand how preprocessing, feature extraction, and classical ML models are applied
- Interpret business outcomes and KPIs derived from NLP solutions
- Reflect on challenges and share domain-relevant examples where similar NLP workflows can be applied

Brief Agenda

Table 1: Brief Agenda

Learning Objective	Comments	Duration	Mode
Part I: Introduction	Overview of session goals, importance of NLP in industry, and session flow	10 mins	-
Part II-A: Case Study 1	Walkthrough of the first NLP use case, covering the problem, approach, and outcomes	25 mins	Presentation
Part II-B: Case Study 2	Walkthrough of the second NLP use case, covering the problem, approach, and outcomes	25 mins	Presentation

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Table 1: Brief Agenda (Continued)

Learning Objective	Comments	Duration	Mode
Part IV: Comparative Insights	Highlight similarities, differences, challenges, and lessons across both case studies	20 mins	Discussion
Part V: Learner Discussion	Guided activity: learners share and brainstorm where similar NLP workflows could be applied in their own domains	15 mins	Interactive
Part VI: Q&A	Address learner questions, clarify concepts, and reflect on industry applications	15 mins	Open Q&A
Part VII: Conclusion & What's Ahead	Wrap-up of learnings and preview of what's next in NLP	10 mins	–

Conclusions

By the end of this live session, learners will have strengthened their understanding of how NLP is applied to real business problems through industry case studies. They will be confident in mapping business challenges to NLP tasks, walking through the traditional NLP pipeline (from preprocessing to model evaluation), and interpreting results in terms of measurable business impact. Learners will also be able to reflect on limitations, compare multiple use cases, and identify opportunities to apply similar NLP workflows in their own domains.

Additional Readings

The following additional links have been provided as references for additional learning:

- [Top 8 Applications of Natural Language Processing \(NLP\)](#)
- [Transforming Industries with NLP: Real-World Examples](#)
- [NLP use cases in the insurance industry](#)